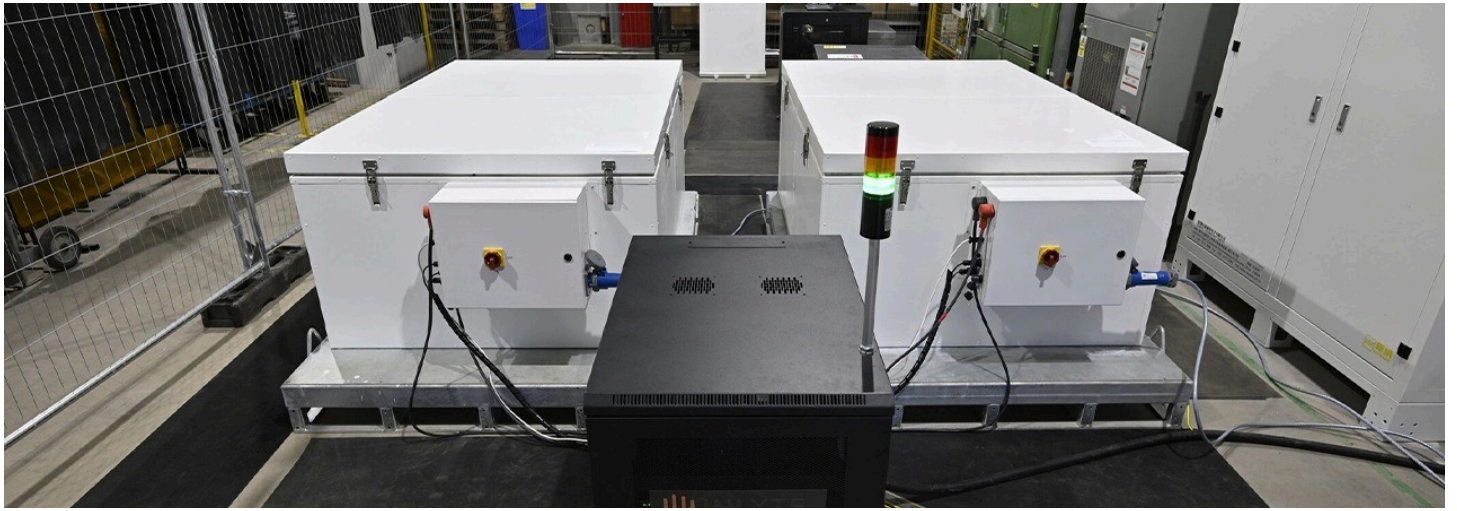




Analysis of Factory Testing for Industrial and Commercial Energy Storage Cabinets



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The tests for the DC side and auxiliary systems are largely similar, but because commercial and industrial energy storage cabinets integrate two additional devices—PCS and EMS—they differ slightly during the power-on commissioning phase.

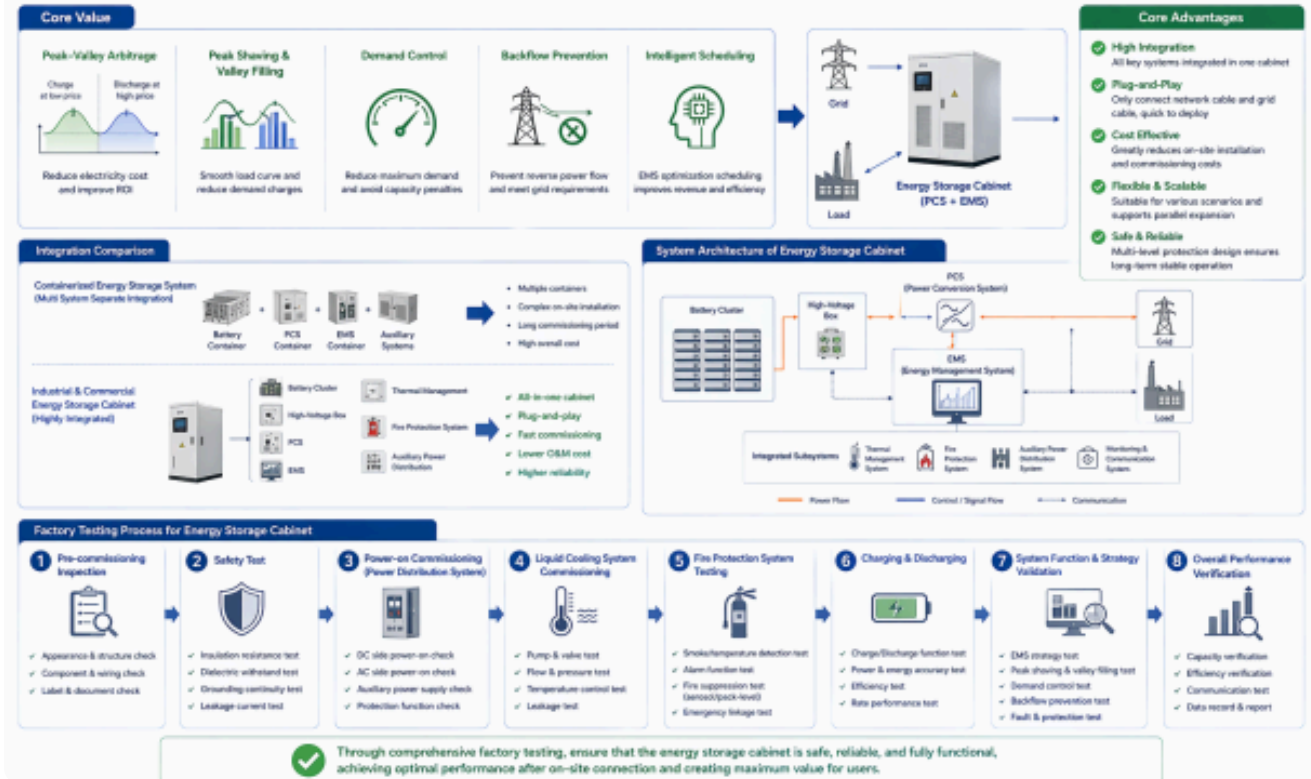
Analysis of Testing Differences between Industrial and Commercial Storage Lockers and Energy Storage Battery Compartments

As the mainstream form of distributed energy storage in industrial and commercial parks, factories, data centers, and commercial complexes, the core value of industrial and commercial energy storage cabinets lies in their ability to achieve peak-valley arbitrage, peak shaving and valley filling, demand control, and backflow prevention through the energy conversion of the PCS and the automatic control strategy scheduling of the EMS. The core advantage is its high degree of integration and plug-and-play capability.

Unlike the "multi system separate integration" of containerized energy storage compartments (where the battery compartment, PCS compartment, and EMS are independently integrated), mainstream industrial and commercial energy storage cabinets integrate battery clusters, high-voltage boxes, PCS, EMS, thermal management systems, fire protection systems, and auxiliary power distribution systems all into a single cabinet. Once transported to the site, they only need to be connected to the network cable and grid connection cable to be put into operation, which greatly reduces the on-site installation and commissioning costs.

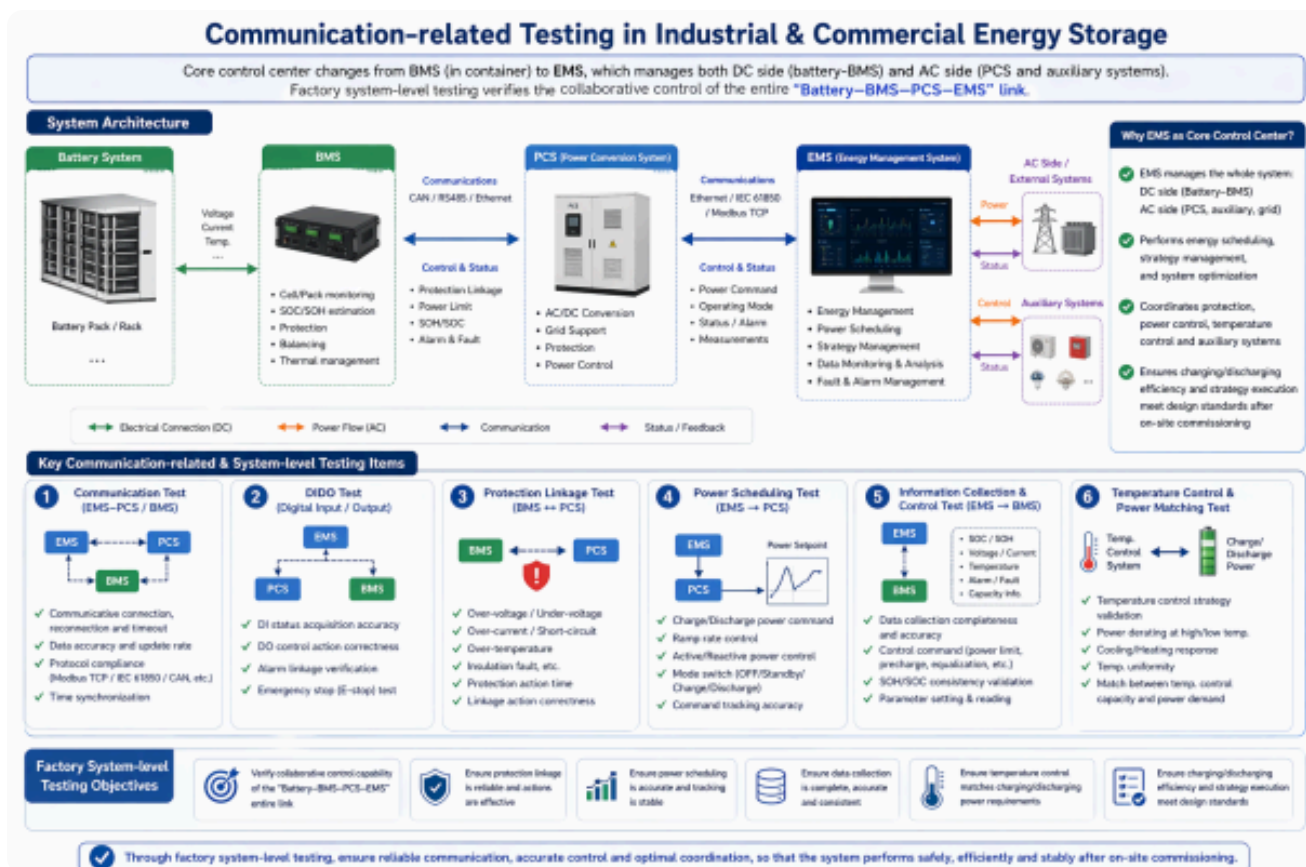
Industrial & Commercial Energy Storage Cabinet: Integrated, Intelligent, Efficient

As the mainstream form of distributed energy storage in industrial parks, factories, data centers and commercial complexes, industrial and commercial energy storage cabinets realize peak-valley arbitrage, peak shaving and valley filling, demand control and backflow prevention through the energy conversion of PCS and the automatic control strategy scheduling of EMS.



Because the cabinet integrates two core components, PCS and EMS, the factory testing of industrial and commercial energy storage cabinets differs somewhat from that of containerized energy storage compartments. Similar to energy storage battery compartments, they also require pre-commissioning checks, safety tests, power-on commissioning of the power distribution system, commissioning of the liquid cooling system, testing of the fire protection system, charging and discharging tests, etc.

However, for communication-related testing, the core control center of the system changes from the BMS in the container to the EMS (in fact, the EMS in industrial and commercial storage is similar to the BAU in the battery compartment, but it manages not only the DC side but also the AC testing equipment). Therefore, in addition to some communication tests and DIDO tests, the system must also verify the collaborative control capability of the entire "battery-BMS-PCS EMS" link.



The protection linkage between BMS and PCS, the power scheduling of EMS to PCS, the information collection and control of EMS to BMS, and the matching of temperature control system and charging and discharging power must all be verified through factory system-level testing to ensure that the charging and discharging efficiency and strategy execution accuracy fully meet the design standards after on-site commissioning.

List of Factory Test Items for Industrial and Commercial Energy Storage Cabinets

The factory testing of industrial and commercial energy storage cabinets, like the overall testing of energy storage modules, follows the principle of static testing before dynamic testing, low-voltage testing before high-voltage testing, individual component testing before system testing, and safety testing before performance testing.

BESS TESTING STAGES & CORE VALIDATION ITEMS

STAGE	TYPE	TEST SUB-PROJECT	CORE POSITIONING
 STATIC POWER-OFF STAGE	 Appearance & Integration Inspection	Cabinet appearance and structure inspection, component installation verification, wiring harness connection and continuity inspection, etc.	 Basic assembly quality verification to eliminate human assembly errors.
	 Electrical Safety Testing	Grounding impedance test, insulation resistance test, withstand voltage test, etc.	 Electrical safety baseline verification to eliminate the risk of high-voltage leakage and breakdown.
 LOW-VOLTAGE POWER-ON STAGE	 Power Supply Verification	After connecting the temporary power supply to the power supply system, perform the switching operations sequentially from high to low and from primary to secondary voltage to check that the voltage meets the equipment's operating range and that the equipment can operate normally.	 Ensure the power supply circuit is safe and reliable, and that the equipment is fault-free.
	 Thermal Management & Fire Protection Testing	Liquid cooling system airtightness inspection, performance testing, etc.; fire detector alarm testing, fire linkage logic verification, etc.	 Temperature control and fire safety verification ensure stable operating environment for battery cells.
	 DIDO Logic Test	After powering on the system circuit, test the system DI points (emergency stop command/AC main circuit breaker status/water immersion/fuse status, etc.) to ensure they can be normally linked with the EMS (or BCU), and the DO points (AC main circuit breaker trip/indicator light/contact, etc.) to operate normally according to the commands.	 Verify that the relevant hard node logic of the system can be executed correctly to ensure system operation.
	 PCS Related Testing	The PCS operating mode, parameter settings, and control strategies have been confirmed to be correct. Communication between the PCS and EMS/BMS, and protection linkage debugging have been performed.	 PCS Function Verification
	 BMS & PCS Integration Testing	BMS cell data acquisition accuracy test, three-level protection logic test, PCS control strategy confirmation, and PCS/EMS linkage protection test.	 BMS Functionality Verification
 SYSTEM INTEGRATION STAGE	 EMS System & End-to-End Communication Testing	Full device communication link testing, EMS core policy logic verification, cloud deployment testing (if applicable).	 The system's "brain" undergoes specialized verification to ensure the accurate execution of scheduling strategies.
 HIGH-VOLTAGE POWER-ON STAGE	 System Cascade Debugging & Charge/Discharge Testing	Commands were sent via EMS to confirm the rapid and correct execution of tasks such as full-charge and discharge tests at rated power for commercial and industrial storage cabinets.	 Verification of system-wide collaborative capabilities to ensure core performance meets standards.
 TEST END	 Reporting & Certification	Organize test content, generate test reports, etc.	 Give the product a certificate of conformity.

⚠️ **NO SHIPMENT BEFORE FAT + MDCC** | FAT: Factory Acceptance Test | MDCC: Manufacturing Data Consistency Check

Comprehensive Analysis of Core Test Items

For industrial and commercial energy storage projects, the relevant testing methods and tools for pre-power-on inspections, fire protection, thermal management, DIDO, etc., are similar to those for large-scale energy storage projects (related insulation and withstand voltage tests should be adjusted according to the actual voltage level to meet national standards).

Preparation phase before system integration testing

1. The system has completed pre-power-on checks, safety tests, system power-on tests, liquid cooling system tests, fire protection system tests, and DIDO tests, and all test results are qualified.
2. The BMS, PCS, and EMS have all completed the latest software upgrades and the strategies have been configured, meeting the operating conditions of the energy storage cabinet.
3. PCS Basic Communication and Hardware Status Testing

No.	Data Point Name	Data Type	Value	Description	Last Updated
1	Module Serial Number	Numeric	5169	A2-SDD4040-BB1	2023-09-21 10:57:33
2	Module Software Version	Numeric	516	2.4	2023-09-21 10:57:33
3	Module Name	Numeric	A2_SDD4040_BB1 	A2_SDD4040_BB1	2023-09-21 10:57:33
4	Module Address	Numeric	3 	3	2023-09-21 10:57:33
5	Total Capacity	Numeric	9600 	9600	2023-09-21 10:57:33
6	Rated Capacity	Numeric	65177 	65177	2023-09-21 10:57:33
7	Calibration Mode	Numeric	0 	None	2023-09-21 10:57:33
8	DO1 Status	Numeric	0 	0	2023-09-21 10:57:33
9	DO2 Status	Numeric	0 	0	2023-09-21 10:57:33


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Testing tools: Laptop, PCS dedicated debugging software, multimeter

Test method: Power on the PCS auxiliary control power supply, establish communication with the PCS through debugging software, read the hardware version and software version, and monitor hardware parameters such as IGBT temperature, DC bus voltage, contactor status, and capacitor voltage in real time.

Acceptance criteria: Stable and uninterrupted communication, hardware version consistent with design, and no alarm messages such as IGBT faults, voltage abnormalities, or communication timeouts.

Core function: To confirm that the PCS's basic communication and hardware status are normal, in preparation for subsequent high-voltage and joint debugging tests.

4. BMS communication and Level 3 protection verification

Testing tools: laptop, CAN box, BMS host computer, etc

Test method:

- Check that the internal communication of the BMS is normal and that the host computer can read all data information from the battery side.
- Ensure that communication between BMS and PCS, EMS, etc., is normal.
- The host computer adjusts the relevant parameters of the three-level protection inside the BMS, and the host computer can read the corresponding alarm status, which is consistent with the parameters.

BMS COMMUNICATION & LEVEL 3 PROTECTION VERIFICATION

Ensuring reliable communication and protection integrity in Battery Energy Storage Systems (BESS)



5. **Acceptance criteria:** Stable and uninterrupted communication, accurate and detailed battery information reading, and normal execution of protection strategies.

6. **Core function:** To ensure that the BMS works properly.

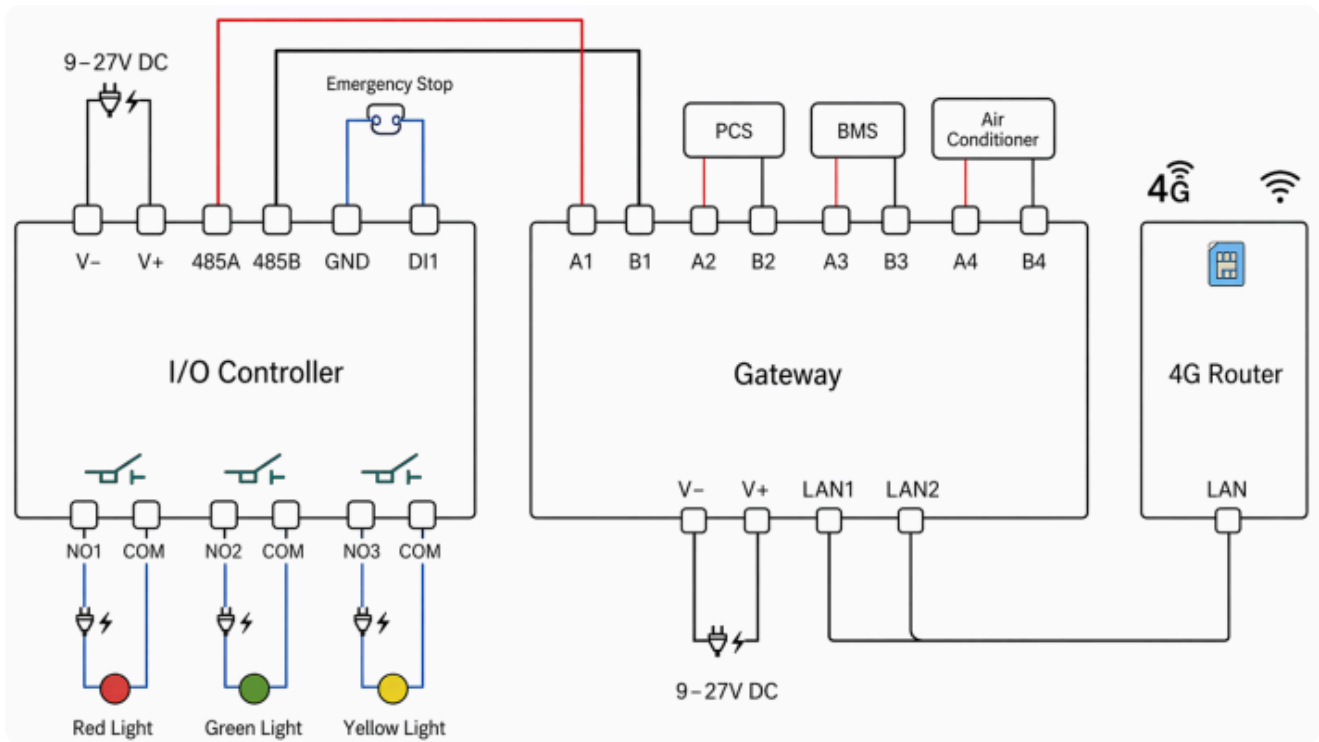
Full device communication link test

Testing tools: Laptop, Modbus debugger, CAN box, network cable tester

Test method: Through the EMS host computer, the end-to-end communication with BMS, PCS, gate electricity meter, liquid chiller, fire alarm control panel and temperature and humidity sensor was tested, covering three communication methods: RS485/CAN/Ethernet, to verify data acquisition accuracy, command issuance response speed and communication stability.

Acceptance criteria: All equipment data is complete and without missing data; analog quantity acquisition cycle $\leq 1s$; digital quantity command issuance response time $\leq 500ms$; continuous 24-hour communication without interruption or data jumps.

Core function: To ensure that EMS can accurately collect data from the entire system and reliably issue control commands, which is the foundation for policy execution.



EMS core strategy logic testing

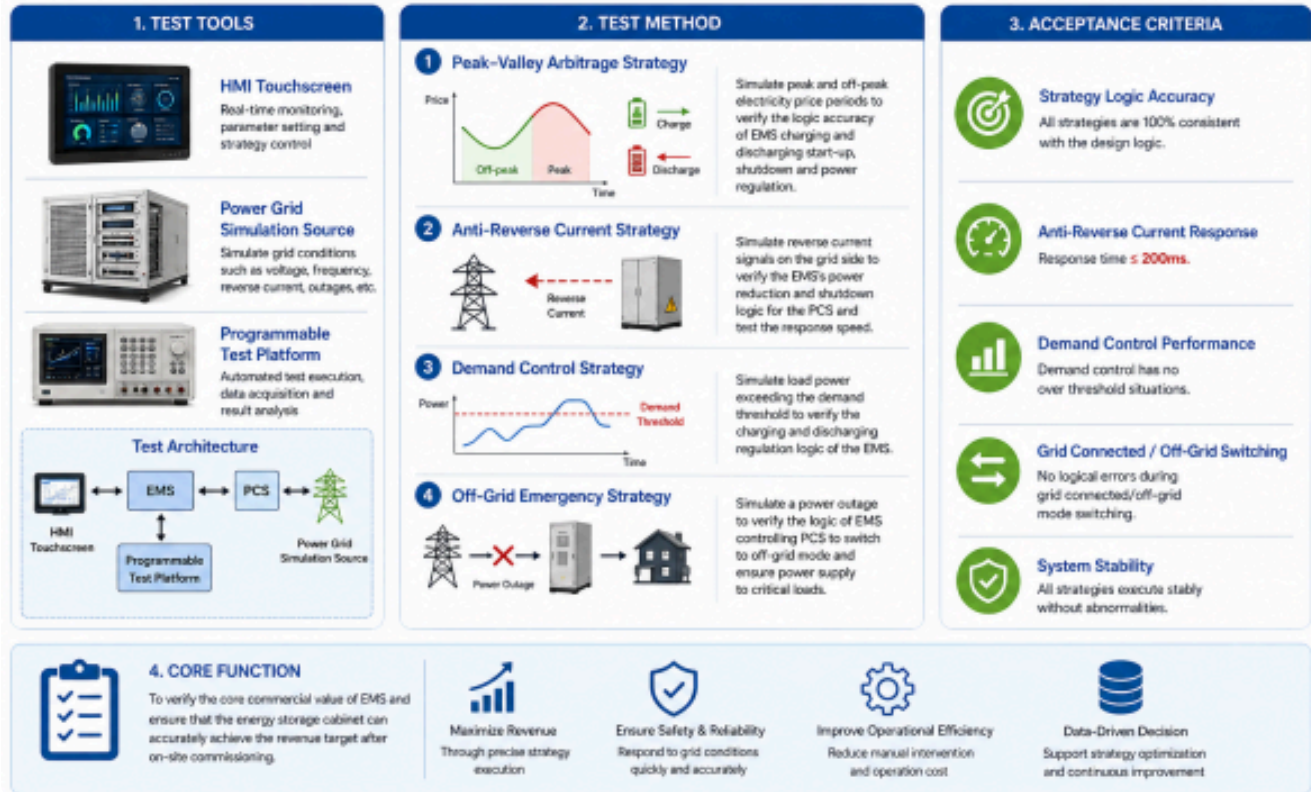
Test tools: HMI touchscreen, power grid simulation source, programmable test platform

Test method:

- Peak-valley arbitrage strategy: Simulate peak and off-peak electricity price periods to verify the logic accuracy of EMS charging and discharging start-up and shutdown and power regulation.
- Anti-reverse current strategy: Simulate reverse current signals on the grid side to verify the EMS's power reduction and shutdown logic for the PCS and test the response speed.
- Demand control strategy: Simulate load power exceeding the demand threshold to verify the charging and discharging regulation logic of the EMS.
- Off-grid emergency strategy: Simulate a power outage to verify the logic of EMS controlling PCS to switch to off-grid mode and ensure power supply to critical loads.

EMS CORE STRATEGY LOGIC TESTING

Verify strategy logic accuracy and ensure energy storage system delivers expected economic value



Acceptance criteria: All strategies are 100% consistent with the design logic; anti-backflow response time is $\leq 200\text{ms}$; demand control has no over threshold situations; and there are no logical errors during grid connected/off-grid switching.

Core function: To verify the core commercial value of EMS and ensure that the energy storage cabinet can accurately achieve the revenue target after on-site commissioning.

Storage Meter Active Power

Please select



Main Transformer Active Power

Please select



Demand Control



Anti-Reverse Flow



Multi-Unit Parallel



SOC Upper Limit

95



SOC Lower Limit

5



Demand Power Upper Limit

Please enter upper limit

Cloud platform integration test

Testing tools: Laptop, cloud platform backend.

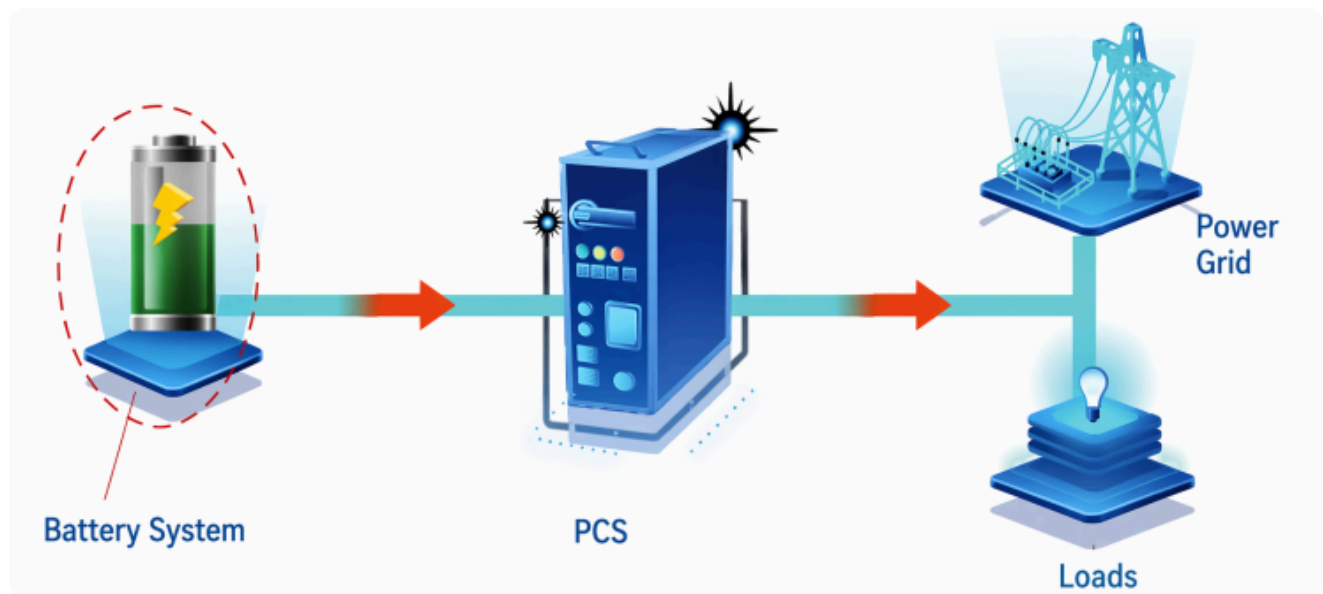
Test methods: Test the storage capacity of EMS local historical data, SOE event logs, and alarm logs, as well as the integrity of data transmission after network outage and data upload to the cloud; test the parameter setting, status viewing, and operation control functions of the local touch screen.

Acceptance criteria: Data is uploaded to the cloud without loss or delay and can be automatically re-uploaded after network outage; the interface operation is smooth, the permission hierarchy is clear, and there are no unauthorized operations.

Core function: To ensure the convenience and safety of on-site operation and maintenance.

System Cascade Debugging and Full Charge/Discharge Process Testing

This is the final verification of the overall system coordination capability of the energy storage cabinet. It must be carried out after all individual tests are passed. The core is to simulate the actual on-site operating scenario and complete the full-process charging and discharging verification.



Testing tools: BMS/EMS/PCS host computer, PC

Test methods: Complete the full power-on process of the system, set the corresponding charging and discharging power and time through EMS, and adjust the charging and discharging power from small to large to the rated power for charging and discharging tests. Run each power for a short time and perform a full charge and discharge test at the rated power (preferably 3 times); monitor the cell voltage, temperature, PCS operating status, and EMS strategy execution in real time, and calculate the system efficiency.

Qualification Standard: The entire charge and discharge process is free of fault alarms and abnormal shutdowns; the charge and discharge power control is accurate, and the system efficiency meets the design requirements.

Core Role: The collaborative operation capability of the entire "battery-BMS-PCS EMS" system was fully verified to ensure that the core performance of the energy storage cabinet fully meets the design standards.

Precautions and Handling of Frequently Asked Questions During Testing

Precautions during the testing process

Commercial and industrial energy storage cabinets are compact in space and densely packed with high-voltage components. Therefore, the following specifications must be strictly adhered to during testing to prevent equipment damage and safety accidents:

Strictly adhere to the testing procedures and strictly prohibit unauthorized operations.

Strict adherence to procedures is required. Proceeding to the next stage is strictly prohibited until all preceding tests are completed and passed. In particular, it is strictly forbidden to apply high voltage to the PCS without performing insulation testing, in order to avoid IGBT breakdown and damage.



High-voltage testing requires proper device protection.

Before insulation and withstand voltage tests, sensitive electronic components such as the PCS control board, surge protector, and EMI filter must be disconnected, and the gate and emitter of the IGBT must be short-circuited to prevent high voltage electrostatic discharge from damaging the power devices.



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Protect Devices
Ensure Safety

PROPER DEVICE PROTECTION – BEFORE HIGH-VOLTAGE TESTING

1

Disconnect sensitive components

Disconnect the following components from the circuit.

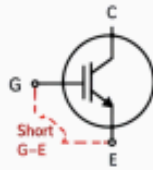
- PCS control board
- Surge protector
- EMI filter



2

Short-circuit the IGBT gate and emitter

Short-circuit the gate (G) and emitter (E) of each IGBT to prevent ESD and protect the device.



3

Perform insulation / withstand voltage test

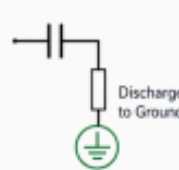
Carry out the insulation resistance test or withstand voltage test according to the test procedure.



4

Remove test voltage and discharge

After the test is completed, remove the high voltage and discharge the circuit completely.



5

Reconnect components and restore

Reconnect all components and confirm correct wiring before power-on.



WHY IS THIS IMPORTANT?

- ✓ High voltage can cause electrostatic discharge (ESD) that may damage control circuits and power devices.
- ✓ Disconnecting sensitive parts and short-circuiting IGBT G-E provides a safe path for ESD, protecting the equipment.



WHAT CAN HAPPEN IF NOT PROTECTED?



Control board damage



Surge protector failure



EMI filter breakdown



IGBT failure



Always follow the test procedure and ensure proper device protection to safeguard equipment and personnel.

The entire charge and discharge test was monitored by dedicated personnel.

The cabinet space is compact, and the risk of thermal runaway of the battery cells is much higher than that of split-type energy storage systems. During the charging and discharging test, a dedicated person must be on duty at all times to monitor the battery cell voltage, temperature and insulation resistance in real time. If any abnormality occurs, the emergency stop button must be pressed immediately to terminate the test.



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During the charging and discharging test, a dedicated person must be on duty at all times to monitor the battery cell voltage, temperature and insulation resistance in real time.

WHY DEDICATED MONITORING IS ESSENTIAL



High energy density in a compact cabinet increases the risk of overheating and thermal runaway.



Real-time monitoring helps detect abnormalities early and prevent accidents.



Dedicated personnel can respond immediately to ensure equipment and personnel safety.

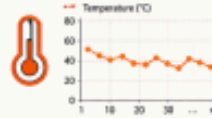
REAL-TIME MONITORING ITEMS

1. Cell Voltage Monitoring



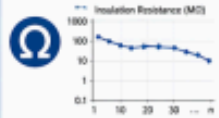
Monitor the voltage of each cell in real time to ensure voltage consistency and detect over-voltage or under-voltage.

2. Temperature Monitoring



Monitor cell and cabinet temperature in real time to detect overheating and prevent thermal runaway.

3. Insulation Resistance Monitoring



Monitor insulation resistance in real time to ensure the system is safe and free from insulation deterioration.

DEDICATED PERSONNEL ON DUTY



IF ANY
ABNORMALITY
OCCURS



1. Detect abnormality

Over-voltage, over-temperature, insulation drop, etc.



2. Press the emergency stop button immediately

Terminate charging/discharging test at once.



3. Ensure system safety

Protect equipment and personnel, and investigate the cause.

Examples of Abnormal Conditions

- Cell voltage exceeds upper/lower limit
- Cell or cabinet temperature exceeds limit
- Insulation resistance drops below threshold
- Communication failure or BMS alarm
- Smoke, odor, or other unexpected events



Dedicated monitoring, timely response, and immediate action are the key to ensuring a safe and successful charge and discharge test.

Fire safety testing must be conducted with appropriate protective measures.

Before conducting a fire alarm linkage test, the solenoid valve of the fire extinguishing device must be removed and the safety pin inserted. It is strictly forbidden to conduct a linkage test with fire extinguishing agents inside. The space inside the cabinet is small, and accidental spraying will cause all electronic components to become damp and damaged, and they will be irreparable.

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Before conducting a fire alarm linkage test, the solenoid valve of the fire extinguishing device **must be removed and the safety pin inserted**. It is **strictly forbidden** to conduct a linkage test with fire extinguishing agents inside. The space inside the cabinet is small, and accidental spraying will cause all electronic components to become damp and damaged, and they will be irreparable.

STRICTLY FORBIDDEN
Do not conduct linkage test with fire extinguishing agents inside the cabinet.

1. BEFORE FIRE ALARM LINKAGE TEST – TAKE PROPER PROTECTIVE MEASURES

2. CONDUCT FIRE ALARM LINKAGE TEST

1 Locate the fire extinguishing device

Find the solenoid valve of the fire extinguishing device in the cabinet.

2 Remove the solenoid valve

Carefully unscrew and remove the solenoid valve.

3 Insert the safety pin

Insert the safety pin into the valve outlet to prevent accidental actuation.

4 Confirm protection in place

Ensure the safety pin is properly inserted and the valve cannot be activated.

3. WHY THIS IS CRITICAL

The cabinet space is compact. Accidental spraying of fire extinguishing agents will cause serious consequences.

Accidental spraying inside the cabinet

All electronic components become damp

Component corrosion, short circuit, and equipment damage

Irreparable damage and long downtime

Proceed with the fire alarm linkage test (such as alarm signal and system linkage verification).

Verify that alarms and linkages function properly.

4. AFTER TEST – RESTORE AND VERIFY

1 Remove the safety pin

Remove the safety pin.

2 Reinstall the solenoid valve

Reinstall and tighten the solenoid valve securely.

3 System check

Ensure the fire extinguishing system is in normal ready status.

SAFETY FIRST • FOLLOW PROCEDURES • PROTECT EQUIPMENT • PREVENT LOSS

Protect People

Protect Equipment

Ensure Reliability

Test data is fully traceable

All test items must be recorded in real time by designated personnel, and standardized test record forms must be filled out. Any non-conforming items must be rectified in a closed loop and retested. All test records, rectification records, and retest records must be archived and retained.

Analysis of Frequently Asked Questions and Standardized Handling Procedures During Testing

Based on actual implementation in industry factory testing, we have identified the six most common problems encountered during the factory testing of industrial and commercial energy storage cabinets, clarifying their phenomena, root causes, and standardized handling methods:

BESS Common Faults – Reasons & Standardized Solutions

Fault Phenomenon	Core Reason Analysis	Standardized Processing Methods
 PCS pre-charge failed, unable to apply high voltage.	1. Precharge resistor damaged; 2. Precharge contactor contacts stuck together; 3. PCS DC bus capacitor damaged; 4. BMS communication with PCS interrupted, unable to issue closing command.	1. Measure resistance values and replace damaged components; 2. Inspect contactors and replace faulty contactors; 3. Inspect bus capacitors and replace damaged capacitors; 4. Repair the communication link between the BMS and PCS, and verify the protocol and address codes.
 EMS anti-backflow strategy failed, backflow exceeded limit.	1. Communication between the meter and EMS is interrupted, resulting in abnormal power grid data acquisition; 2. Incorrect meter address and power direction configuration leads to reverse data acquisition; 3. The EMS anti-reverse current response cycle is set too long, causing power regulation lag; 4. Delay in EMS and PCS command interaction.	1. Verify the communication wiring, address, and protocol of the electricity meters, repair the communication link, and ensure real-time acquisition of grid power data; 2. Correct the power direction configuration of the electricity meters to ensure accurate acquisition of reverse flow data; 3. Optimize EMS strategy parameters, shorten the response cycle, and set a reasonable power adjustment step size; 4. Optimize the communication cycle between EMS and PCS to ensure zero delay in command issuance.
 Excessive temperature difference of battery cells during charging and discharging.	1. Uneven flow distribution in the liquid cooling pipeline, resulting in insufficient coolant flow in some packs; 2. Mismatch between the liquid chiller's temperature control strategy and charging/discharging power, leading to lag in water temperature regulation; 3. Temperature sensor acquisition deviation.	1. Adjust the balancing valve in the liquid cooling pipeline to ensure uniform flow in each pack and clear any blockages in the pipeline; 2. Optimize the linkage temperature control strategy between the liquid chiller and EMS, adjusting the water temperature in advance according to the charging and discharging power; 4. Calibrate the temperature sensor and correct any data acquisition deviations.
 All devices experienced communication outages, and EMS data went offline.	1. Power supply failure of the switch inside the cabinet, or incorrect network cable wiring; 2. IP address or address code conflicts among multiple devices, causing bus paralysis; 3. Communication lines and power lines are laid in the same trench, causing strong electromagnetic interference and communication interruption; 4. EMS controller failure.	1. Check the power supply and network cable continuity of the switch, and replace the faulty switch; 2. Verify the IP addresses and address codes of all devices to ensure there are no conflicts; 3. Lay communication lines and power lines separately, with a parallel spacing of $\geq 20\text{cm}$, replace shielded communication cables, and ground the shielding layer at only one end; 4. Check the hardware status of the EMS controller and replace the faulty equipment.

Note: After troubleshooting, perform system verification to ensure the fault is completely resolved and the system operates normally.

Conclusion

The factory testing of industrial and commercial energy storage cabinets verifies not only the performance of individual components but also the coordinated control capabilities of the entire system. Only by completing the entire process of standardized testing in the factory can industrial and commercial energy storage cabinets truly achieve "plug and play," enabling smooth grid connection and stable operation after the equipment arrives on site, and accurately realizing the commercial value of peak-valley arbitrage and demand control.

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